



Sharif University of Technology
Department of Computer Engineering

B.Sc. Thesis

Self-Supervised Cell Segmentation in Microscopy Images

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In the name of God

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Abstract

With the widespread progress in imaging tissue cells, identifying and diagnosing complex diseases with machine learning methods has become very important. Cell characteristics such as size and shape are important factors in determining cell type, and they can be learned with methods such as training a deep neural network. In this project, we aim to segment cells in microscopy images with a self-supervised method. The goal of segmenting these images is to detect the nucleus in different types of cells. The image data used for nucleus detection is the Kaggle Data Science Bowl 2018 dataset, which is publicly available.

Keywords: self-supervised learning, cell segmentation, nucleus detection, neural network

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Chapter 1

Introduction

This chapter presents the importance of the topic, the problem definition and related literature, the research objectives, and the structure of the thesis.

1.1 Importance of the Topic

Segmenting cell nuclei not only gives us information about their location and density, it also provides many morphological features (such as size and nucleus-to-cytoplasm ratio) relevant to tumour diagnosis and treatment procedures [2]. For this reason, a great deal of research has been devoted to building automatic systems for accurate nucleus segmentation.

1.2 Problem Definition and Related Literature

Although deep-learning approaches have achieved outstanding segmentation accuracy on nucleus samples, the scarcity of annotated data is a common obstacle to further progress. Because nuclei are widespread and show large morphological variation, annotating nucleus instances requires experienced physicians to review histopathology images repeatedly and trace their outlines precisely, which is very difficult and costly. As a result, the performance of deep-learning approaches suffers from the small number of annotated histopathology images. Automatic localisation of nuclei is now within reach with deep learning and machine learning, which is why Kaggle launched the Kaggle Data Science Bowl 2018 competition, leading to new solutions for finding cell nuclei. Kaggle also released the competition data publicly so that the best solution could be found.

Self-supervised learning, as a way to remove the need for manual annotation when training neural networks, has attracted growing attention. The pipeline usually consists of two stages:

1. Pretrain the network on a *proxy task* using a large unlabeled dataset.

2. Fine-tune the pretrained network for the specific target task using a small annotated dataset.

Recent studies have confirmed the effectiveness of self-supervised learning on various medical image processing tasks, such as brain-region segmentation, brain-tumour segmentation and organ segmentation [3]. However, only a few studies have focused on instance segmentation of cell nuclei, which is quite different from semantic segmentation [4]: a network performing instance segmentation must not only classify objects, but also identify, for every pixel, the object instance it belongs to.

1.3 Research Objectives

In this thesis, a self-supervised learning framework is proposed and implemented to exploit the capacity of deep convolutional neural networks for nucleus segmentation. The proposed self-supervised proxy task consists of two sub-tasks, *scale-wise triplet learning* and *count ranking*, which lead the network to learn prior knowledge about nucleus size and quantity automatically and independently, by making deep use of the rich information in the raw data.

1.4 Thesis Structure

This thesis consists of six chapters. Chapter 2 covers the background concepts related to the thesis. Chapter 3 reviews previous work in this area. Chapter 4 presents our proposed model for the problem. Chapter 5 describes the experiments designed to compare our model with previous models, and also suggests directions for future work. Chapter 6 concludes the thesis.

Chapter 2

Background

2.1 Self-Supervised Learning

Self-supervised learning is a way to train computers to perform tasks without humans providing labeled data. It is a subset of unsupervised learning in which the outputs or goals are obtained by machines that label, categorise and analyse information on their own, and then draw conclusions based on relationships and correlations.

Self-supervised learning allows AI systems, despite the scarcity of labeled datasets, to break complex tasks down into simpler ones in order to reach a desired output. The main idea is to automatically generate a kind of supervisory signal for solving tasks; therefore, in order to predict the output, the model must extract good patterns from the data and build a good representation of the process.

Producing a labeled dataset is very expensive, especially for medical images, which are abundant and must be labeled (annotated) by experts, while unlabeled data is being produced in large quantities. One way to use this wealth of unlabeled information is to set the learning objectives correctly so that the supervision comes from the data itself.

One self-supervised learning task is to extract several patches from an image and have the model predict the relationships between the patches.

2.2 The U-Net Neural Network

U-Net is a deep neural network designed in 2015 for segmenting medical images. The main idea of a convolutional neural network is to learn the feature maps of an image and use them to build more refined feature maps. This works well for classification problems, because the image is converted into a vector that is then used for classification. In image segmentation, however, we must not only convert the feature map into a vector but also reconstruct an image from this vector. This is the core idea behind U-Net.

U-Net uses the same feature maps that were used for contraction to expand the vector

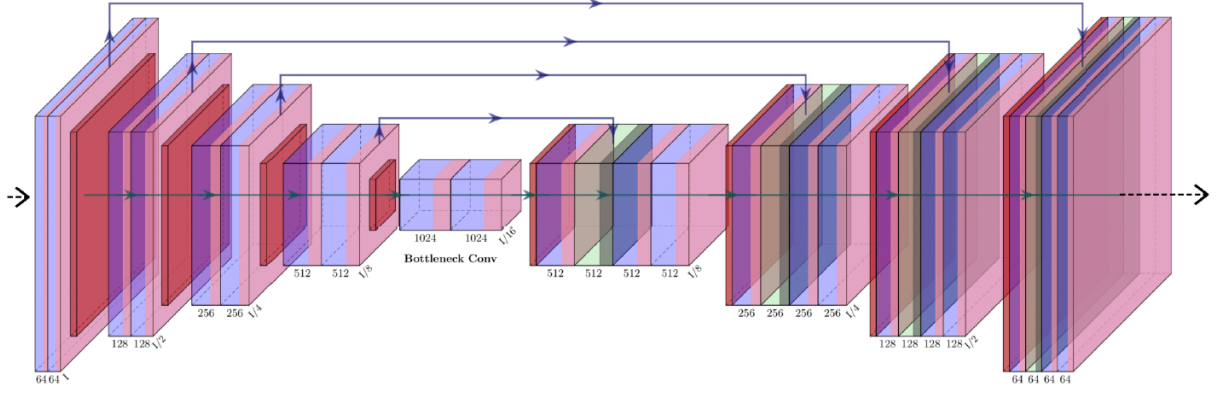


Figure 2.1: The U-Net model used.

back into a segmented image. The architecture is U-shaped and consists of three parts: the contraction path, the bottleneck and the expansion path. The number of expansion blocks equals the number of contraction blocks [5].

The network used in this method is the U-Net from [6], which was used there to solve jigsaw puzzles. This model, shown schematically in Figure 2.1, is an autoencoder whose down-sampling (contracting) part is used as the encoder and whose up-sampling part is used as the decoder.

Chapter 3

Previous Work

A large part of the literature on medical images concerns localising and finding cell nuclei. In this chapter we review the work of the top teams in the Kaggle Data Science Bowl 2018 competition.

- **First team.** This team used an ensemble of models together with data augmentation and a post-processing stage. They used 8 fully convolutional networks and repeated this part of the model 4 times, training 32 networks in total. The backbone of all 8 networks is an encoder-decoder: 6 of them use a U-Net and the other two form FPN networks. The encoders are ResNet networks. The team followed two strategies to avoid overfitting: first, using networks pretrained on the ImageNet dataset, and second, generating and augmenting data with 24 different methods.
- **Second team.** This team used an FPN-based network whose last two layers were customised. The mask location is obtained relative to the cell position from 4 directions (vertical, horizontal, 45 degrees and 135 degrees). The resulting mask then goes through post-processing and is turned into the cell region. This FPN was also pretrained on ImageNet. The team also introduced a new cost function that reduces the error rate for object size, effectively weighting small and large objects equally. This team also used data augmentation during training and testing.
- **Third team.** This team used a Mask R-CNN model pretrained on the COCO dataset. It also used scale-based data augmentation (shrinking and enlarging images); mirroring and rotating images were among its other augmentations. It also used test-time augmentation [7].

Chapter 4

Proposed Method

This chapter describes the proposed self-supervised proxy task and its details.

4.1 Triplet Dataset

Triplet learning methods, whose goal is to learn an embedding space in which similarities between data points are preserved, are used to encourage the neural network to implicitly learn features of nucleus instances (such as the size and number of nuclei). In our triplet learning method, we take a random patch of a given size from the image as the *anchor*; then, with a small random *shift* of 0 to 50 pixels, we create the *positive* patch. The positive image is expected to be similar to the original image in scale and number of cells. To better learn nucleus-size information in the self-supervised method, a *negative* sample containing larger nuclei is produced: a smaller patch is randomly cropped from the positive sample and resized to the size of the positive image. The negative image is expected to have a larger scale than the anchor and positive images, and fewer cells than the positive image.

Since we worked with two datasets, for MoNuSeg, whose images are 1000×1000 pixels, we follow [4] and set the size of the anchor and positive samples to 768×768 . To create more variety, the negative crop is taken from the positive image at sizes 512×512 , 256×256 , 128×128 and 64×64 , and the cropped image is enlarged to 768×768 to create the negative sample. For the Kaggle dataset, whose images are 512×512 , the anchor and positive samples are 384×384 and the negative crops are taken at 256×256 , 128×128 and 64×64 . The anchor, positive and negative samples form a standard triplet data point, which is used for the proxy task in self-supervised learning.

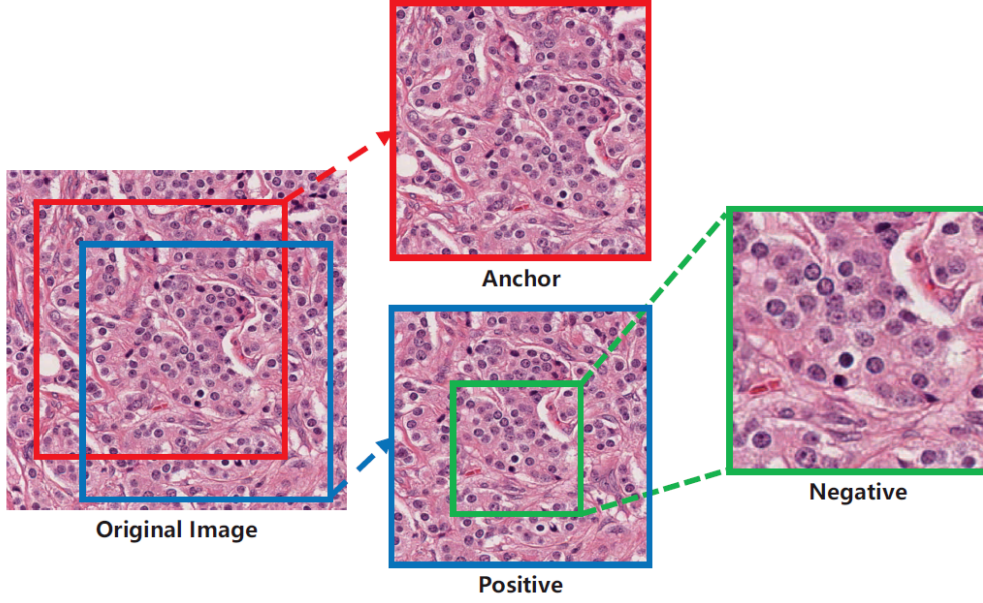


Figure 4.1: Anchor (red crop), positive (blue crop) and negative (green crop) samples in the MoNuSeg dataset.

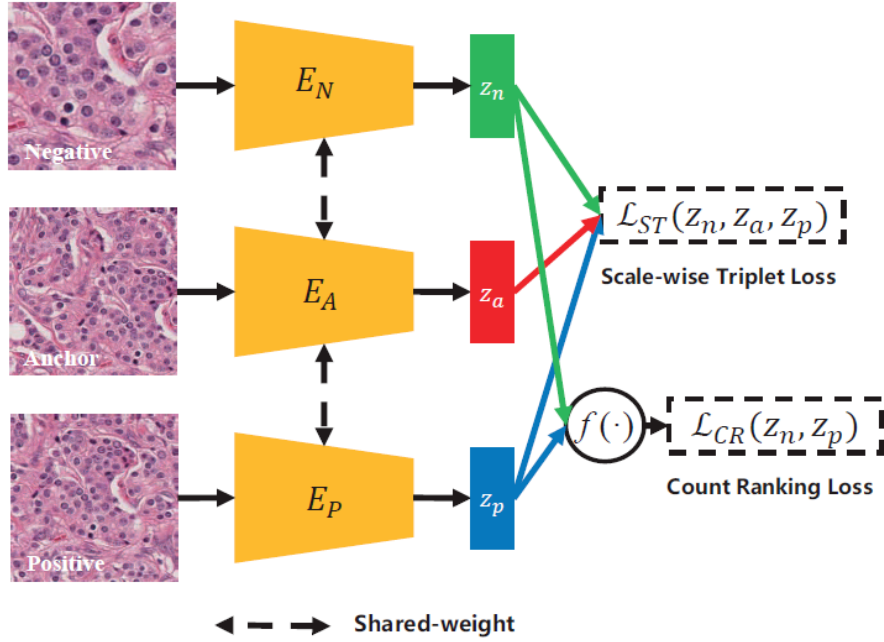


Figure 4.2: Encoder model with shared weights.

4.2 Self-Supervised Approach and Proxy Tasks

As mentioned, to pretrain the neural network we use two proxy tasks: triplet learning and count ranking. To implement this, we use three encoders with shared weights (Figure 4.2), which serve two purposes designed to learn image features. The first task, scale-wise triplet learning, relates to recognising size and scale; the second, count ranking, relates to the number of cells in the image.

The cost functions in this part are defined as follows.

4.2.1 Scale-wise Triplet Learning

Here the model’s output is first mapped linearly to (z_a, z_p, z_n) , and a cost function for the effect of scale, defined below, is applied to them [1]:

$$L_{ST}(z_a, z_p, z_n) = \sum \max(0, d(z_a, z_p) - d(z_a, z_n) + m_1) \quad (4.1)$$

In this formula, m_1 is the *margin* (set empirically to 1), and d is the PairwiseDistance function, which computes the batch-wise distance between its two inputs using the p -norm. With this triplet loss, the network learns to reduce the perceptual distance between the anchor and positive samples in the feature space, and to increase the dissimilarity in meaning (such as nucleus size) between the anchor and negative samples.

4.2.2 Count Ranking

This cost function is applied only to the positive and negative samples, to capture the effect of the number of cells in the images. As in the previous formula, m_2 is the margin (set empirically to 1). The function f is implemented by a fully convolutional layer.

$$L_{CR}(z_p, z_n) = \sum \max(0, f(z_n) - f(z_p) + m_2) \quad (4.2)$$

Finally, using the sum of the two functions above, the loss of the self-supervised model is computed as:

$$L = L_{ST} + L_{CR} \quad (4.3)$$

4.3 Fine-Tuning on the Main Task

The main task here is to classify each pixel of the image into three categories: background, cell body and cell boundary. We achieve this by taking the trained encoder and connecting it to a decoder whose weights are randomly initialised. The reference paper [4] used a ResNet-101 network as the encoder. We first tested this network, but because of its

complexity and large size it was replaced with the network presented in [6]. This model is a U-shaped U-Net. We use its down-sampling part as the encoder, apply the tasks above to it and pretrain it so that it extracts sample features and updates the network weights. We then complete the U-Net with these updated weights and initialise its decoder randomly. Finally, the network is trained to segment cells on annotated data with the embedded Binary Cross Entropy loss (`BCEWithLogitsLoss`).

Chapter 5

Experiments and Results

5.1 Datasets

5.1.1 MoNuSeg 2018 Dataset

The dataset of this challenge was obtained by carefully annotating tissue images of several patients with tumours of different organs (including liver, kidney, breast, prostate, bladder, colon and stomach), diagnosed in 18 hospitals. It contains diverse images with H&E staining at $40\times$ magnification. H&E staining is a common protocol for enhancing the contrast of a tissue section and is typically used to assess tumours (grading, staging, etc.).

The dataset has a training set of 30 manually annotated images and a test set of 14 images. The resolution of the histopathology images is 1000×1000 pixels. The annotations of each image are stored in an XML file containing the exact segmentation of that image [4].

5.1.2 Kaggle Data Science Bowl 2018 Dataset

These images form a diverse set of biological images containing tens of thousands of nuclei in total. The diversity of the datasets reflects the types of images collected by biologists at universities, biotech companies and hospitals. The nuclei come from a wide range of organisms, including humans, mice and flies. In addition, the nuclei were imaged under different conditions, including fluorescent and histological stains, several magnifications and varying lighting quality. Finally, the nuclei appear in different contexts and states, including cultured monolayers, tissues and embryos, cell division and genotoxic stress. The dataset was designed to challenge an algorithm’s ability to generalise across these variations.

The dataset has two parts (Stage1 and Stage2), of which we used Stage1. For uniformity, we split the images into patches of 512×512 pixels. The Stage1 training set contains

1,162 annotated images and its test set contains 126 annotated images. The annotations are provided as a set of masks for each image; each mask is a PNG file containing the exact segmentation of that image [8].

5.2 Experimental Setup

To train the network, in the first stage we convert the Stage1 training data into triplets of anchor, positive and negative samples and feed this training set to the encoder part of the network (this stage uses only the raw images and does not need annotated images). After this stage is complete, we transfer the network weights to the next stage. In this stage we use the complete U-Net, and to train it we use 10% of the annotated Stage1 training data. Finally, we test the trained U-Net on the Stage1 test data.

5.3 Evaluation and Performance

To evaluate the accuracy of our model, we use the same evaluation metric as the Kaggle Data Science Bowl 2018 competition, Mean Average Precision, so that we can compare our results with the competition’s participants. This strategy is based on object-level detection errors. Participants identified the target masks and were ranked by matching these masks with the ground-truth masks and computing TP (true positives) and FP (false positives). The match between two masks is measured by the IoU, computed as:

$$IoU = \frac{|A \cap B|}{|A \cup B|} \quad (5.1)$$

where A and B are the two objects (target and prediction) and $|\cdot|$ denotes area. To compute the participants’ scores, a threshold is chosen, and matches below this threshold are counted as errors. Using the IoU, precision, recall and the F1 score are computed as follows:

$$P(t) = \frac{TP(t)}{TP(t) + FP(t)} \quad (5.2)$$

$$R(t) = \frac{TP(t)}{TP(t) + FN(t)} \quad (5.3)$$

$$F1(t) = \frac{2TP(t)}{2TP(t) + FP(t) + FN(t)} \quad (5.4)$$

If a team detects the target mask exactly, the IoU equals 1, which is the best case. In practice this is not possible, and only part of the target mask can be covered. For this

reason, for comparison over increasing IoU thresholds, a score is computed as:

$$S = \frac{1}{|T|} \sum_{t \in T} \frac{TP(t)}{TP(t) + FP(t) + FN(t)}, \quad T = 0.10, 0.15, \dots, 0.95 \quad (5.5)$$

Based on this metric, we trained and evaluated our model for different numbers of epochs. To assess our method, we also trained exactly the same U-Net for these numbers of epochs without the self-supervised method, using the same 10% of the training data, tested it on the test data, and evaluated that model with the same metric. The results of these evaluations are reported in Table 5.1.

Model	300 epochs				500 epochs			
	Train Loss	Validation Loss	mAP	Test mAP	Train Loss	Validation Loss	mAP	Test mAP
U-Net from scratch	0.0956	0.1300	0.4669	0.2608	0.0771	0.1223	0.4923	0.2608
Self-supervised (10 epochs)	0.0929	0.1266	0.4450	0.2655	0.0807	0.1235	0.4803	0.2545
Self-supervised (30 epochs)	0.0914	0.1245	0.4400	0.2697	0.0789	0.1301	0.5047	0.2735

Table 5.1: Evaluation results. “Self-supervised (n epochs)” denotes n epochs of self-supervised pretraining of the encoder before training the full U-Net.

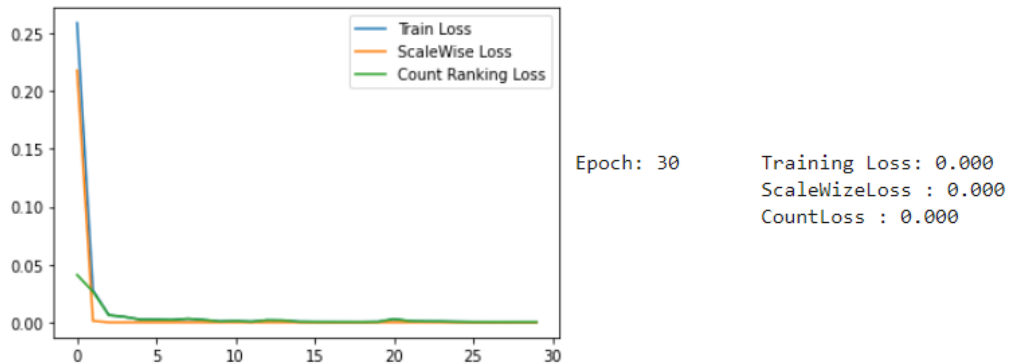


Figure 5.1: Loss curves of the self-supervised part of the model.

Finally, with 30 epochs for the self-supervised part and 500 epochs for training the whole network, the model reached its best result: $mAP = 0.5047$ on the validation data and $mAP = 0.2735$ on the test data.

5.4 Suggestions for Improving the Model in Future Work

Examining the results and the plots of the designed model, we find that the model overfits in the self-supervised part, which is also why the difference from the non-self-supervised

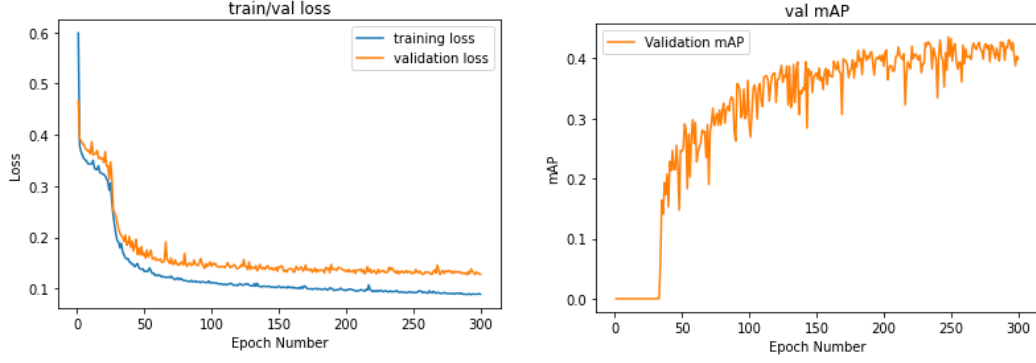


Figure 5.2: Results of 30 epochs of encoder training and 300 epochs of U-Net training, $mAP = 0.44$.

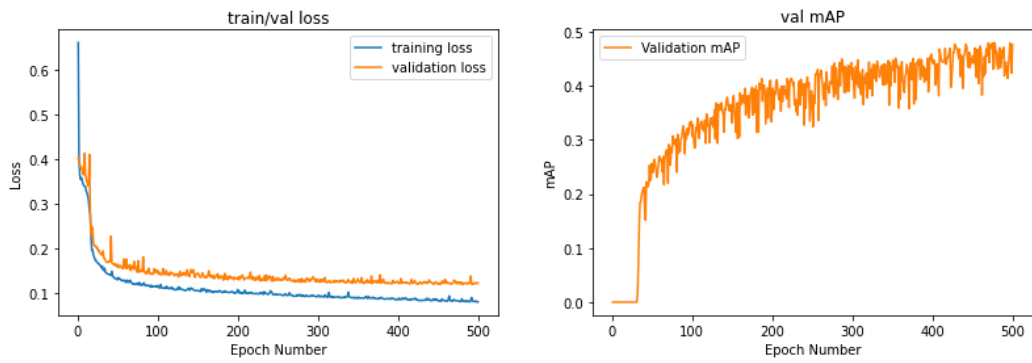


Figure 5.3: Results of 10 epochs of encoder training and 500 epochs of U-Net training, $mAP = 0.48$.

model is small. To improve the performance of this model, we plan in future work to change our loss functions (for example, using softmax instead of max) and to train the encoder network with them. In addition, to prevent the self-supervised network from overfitting, we plan to apply adversarial attacks to the data.

Chapter 6

Conclusion

In this thesis, a framework for cell segmentation with a self-supervised method was presented and implemented. The proposed method consists of two sub-tasks, scale-wise triplet learning and count ranking, which enable the neural network to implicitly acquire knowledge about the size and quantity of nuclei. This self-supervised learning model was evaluated on the publicly available Kaggle dataset and achieved $mAP = 0.5047$.

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